

Perimeter Runs and Misdirection: Jet Sweeps, Reverses, and the Wing-T's Children

How offenses win the race to the sideline, and how they punish a defense that runs too hard to win it back

Third quarter, second-and-8. For the fourth time today a receiver sprints across the formation in jet motion just as the ball is snapped. The first time he got the ball and ran around the left end for eleven yards. The next two times the quarterback stuck the ball in his belly and pulled it out, and both linebackers took three hard steps toward him anyway, while the running back went the other way. This time both linebackers sprint left again, the safety on that side races up to meet the receiver at the corner, and the quarterback hands the ball to the running back going right, into a hole where a linebacker stood a moment ago. Nobody on the left side of the field touched the ball, and every one of them helped make the play. This chapter is about why the defense keeps chasing a receiver who mostly doesn't get the ball, and what happens to defenses that stop.

You'll be able to:

- Explain why runs to the edge are a race, and name the defender each one is trying to outrun or wall off.
- Recognize the jet sweep, the end-around, the reverse, the crack-toss and the buck sweep, and say what each one punishes.
- Spot the two perimeter blocks (the **crack** and the **seal**) and say why a receiver blocking back toward the ball is one of the most effective blocks in football.
- Explain the Wing-T's series logic (buck sweep, trap, waggle, counter) and point to its NFL descendants on a Sunday broadcast.
- Watch the linebackers on jet motion and say whether the motion did its job, even when the ball went somewhere else.
- Explain why a play that gains a lot when it's run rarely can't simply be run more often.

You'll need

- **Gap Schemes:** a **pull** is a lineman leaving his spot at the snap to block somewhere else; a puller either **kicks out** the end man (hits him from the inside and pushes him toward the sideline) or **logs** him (gets outside him and pins him in). The toss is a pitch to a back already running wide, and pin-and-pull is the modern outside run in which the linemen nearest the play block down (pin) and the ones farther away pull around them. Trap lets a defender in and hits him from the side with a pulling lineman; counter starts the backfield one way and pulls blockers back the other.
- **Motion and Shifts:** **jet motion** is a receiver sprinting flat across the formation with the snap timed just before he reaches the quarterback, and fly motion is the Wing-T's name for the same thing. Against motion the defense can *travel* (one defender follows), *bump* (everyone slides a spot) or *spin* (a safety comes down toward the motion and the other goes to the middle).
- From earlier chapters: play side and backside (the half of the formation the run is aimed at, and the other half), the technique numbers used to label defensive linemen (a 5 is outside a tackle, a 9 outside a tight end), the Wing-T formation (a wingback just outside and behind the tight end), and the condensed formation (receivers lined up close to the line, where they can block back inside).

How to read the diagrams. Offense is blue circles, defense orange squares; *left* and *right* are always the offense's. LT, LG, C, RG and RT are the five offensive linemen, Q the quarterback, R the running back, Y the tight end, and X, Z and H receivers (each caption says where they line up). In the Wing-T pictures F is the fullback, H the halfback and W the wingback. Defensive linemen are labelled by technique number; W and M are the Will and Mike linebackers, S the Sam, NB the nickel back, CB a cornerback, FS and SS the safeties. Lines ending in a short bar are blocks, arrows are paths, the thick dark arrow is the ball carrier, and dashed orange arrows are how the defense reacts.

The edge is a race

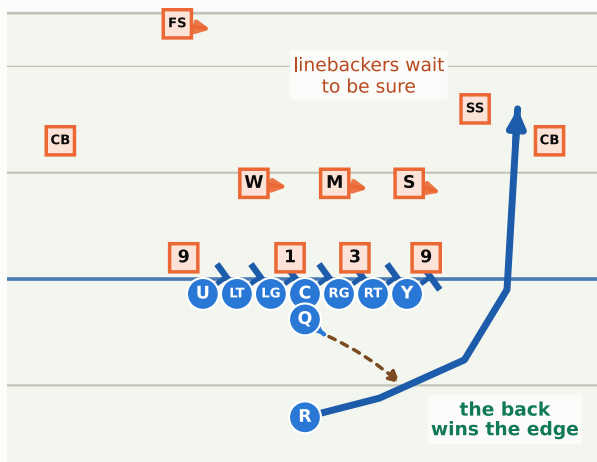
Everything you have seen in the run game so far has been a fight in a phone booth. On inside zone or power, blockers and defenders start a yard apart, and the run hits before the defense can move very far. Runs to the **perimeter**, the edge of the formation and the open grass outside it, are a different contest. The ball carrier has 10 or 15 yards of sideways running to do before he can turn

upfield, and every defender on the field has time to see where he is going and run there too. An outside run is a race between the ball and the defense's **pursuit**, the whole team running to the ball.

The defense organizes that race with two jobs. One defender on each side is the **force** player (more on him in Run Fits): his job is to get outside the ball and turn it back inside, toward everyone else. Everyone else pursues, each on an angle aimed at where the ball carrier *will be*, not where he is. If the force player holds and the pursuit is fast, an outside run dies for no gain or a loss, and losing yards behind the line is the outside run's signature failure.

So the defense wants to pursue *fast*. And that is the opening for everything else in this chapter. A defense that flows hard to stop the sweep leaves the other side of the field empty and its linebackers a step out of place. The offense can punish that in two ways: run the ball back where the defense came from, or simply make the defense *think* the ball is going somewhere it isn't.

Defense reacts slowly



Defense flies to the ball

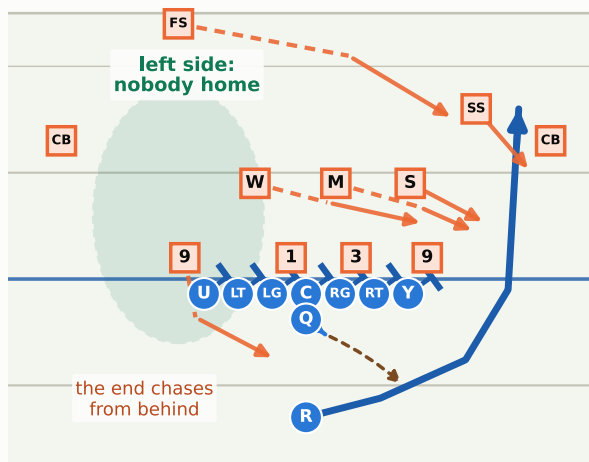


Figure 1: The defense's dilemma on an outside run to the right (here a toss from under center). Left: if the defenders read and react slowly, the back gets outside the end before the linebackers and the safety arrive, and the race is lost. Right: if they all sprint to the ball at the snap (dashed orange arrows), they win the race to the edge, but the left side of the field is now empty (green shading) and the linebackers have run themselves out of the middle. Plays that go back that way, or fake one way and go the other, exist to punish the fast defense.

That picture is the whole chapter in miniature. **Perimeter runs** punish a defense that is slow to the edge. **Misdirection**, any play design that gets the defense to commit in one direction while the ball goes another, punishes a defense that is fast to it. The two families need each other: misdirection only works if the defense has a reason to run hard, and the only reason it has is that the perimeter run will beat it if it doesn't.

The perimeter blocks: crack and seal

Outside runs need two kinds of block that you haven't met yet, because inside runs rarely need them.

A **crack block** is a block thrown by a player lined up *outside* the play, usually a wide receiver or a wingback, who comes back *toward the ball* and hits a defender from the side. The usual target is a linebacker or safety who is watching the ball carrier and running to the edge, which is why the block is so effective: the defender's eyes are on the ball, his momentum is going outside, and the blocker arrives from a direction he isn't looking. The name comes from what it can feel like to the man who gets hit. You met the idea in 02-04, where a condensed split put the receiver close enough to crack in a step or two.

A **seal block** is any block whose job is to *wall a defender off* from the ball's path, rather than to move him. On an outside run that usually means pinning an inside defender inside: the tight end steps down onto the end lined up over him, gets his body between that end and the sideline, and the back runs outside the wall. A seal doesn't need to win the defender's feet back a yard; it only needs to stay in the way while the ball goes by. Coaches use the word loosely (a log block by a puller who gets outside a defender and pins him in is a kind of seal), but the idea is always the same: *you don't move him, you shut him off*.

The two blocks together make the classic outside run of the modern NFL, the **crack-toss**. The receiver cracks the defender who was supposed to force the play, the tight end seals the end, and a lineman pulls around both of them to block the cornerback, the man the receiver would normally have blocked. Coaches call that last part *crack-and-replace*: the receiver leaves his usual man to crack inside, and a puller replaces him.

Crack-toss: the crack, the seal and the replace

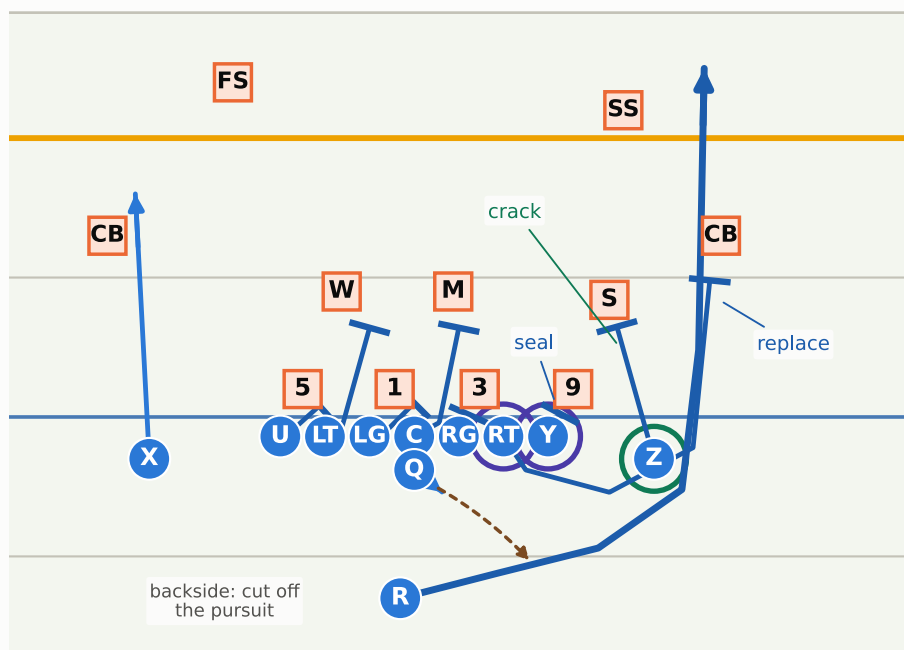


Figure 2: Crack-toss right from a condensed two-tight-end set (12 personnel; Z lines up only about 8½ yards from the ball, a yard off the line). The quarterback tosses to the back (R) heading right. Z cracks (green ring) back inside on the Sam linebacker (S), who is aligned to force the play. The tight end Y seals the 9-technique end, keeping him inside. The right tackle pulls around both blocks and replaces Z on the cornerback. The back’s lane is between the crack and the pulling tackle; the safety (SS) is the defender left over. Backside, the linemen and the second tight end (U) cut off pursuit.

Read Figure 2 as a list of problems and answers. The Sam linebacker lined up outside the tight end is the defense’s force player: if he gets outside the ball, the toss is dead. Nobody on the line can reach him in time, but Z, lined up just outside him, can hit him from the side before he gets going. The 9-technique end is lined up outside the tight end, so the tight end can’t reach around him; instead Y gets his body on the end’s outside shoulder and keeps him from chasing. The cornerback is now unblocked, so the tackle, who has the shortest path of any lineman to the outside, pulls around and blocks him. Every block is an angle the defender wasn’t expecting.

The cost is that crack-toss is fragile in two specific ways. If the cracked defender sees it coming, he can duck under the crack and be in the backfield, and the pulling tackle has to choose between him and the corner. And the cornerback has an answer of his own, which you’ll see in the drills: when he sees the receiver go inside, he yells “crack!” and comes up to take the force job himself, which leaves the puller blocking air and the back running into the man who just replaced the linebacker.

¹NFL Rule 12, Section 2, Article 6 (crackback block; 15 yards), listed in the NFL’s 2025 rulebook penalty table: <https://www.nfl.com/news/2025-rulebook-part-2>. The rule’s three conditions (within five yards of the line of scrimmage; moving toward the spot of the snap; aligned more than two yards outside the tackle, or in motion from the backfield) and the ban on contact below the waist or to the head and neck are summarized at <https://www.footballnationusa.com/post/what-is-a-crackback-block-in-football-nfl-rule-explained>.

²NFL, “Health and safety-related rules changes since 2002” (2019: a blindside block, “a block in which [the blocker] is moving toward or parallel to his own end line and makes forcible contact to his opponent with his helmet, forearm, or shoulder,” became a foul): <https://www.nfl.com/playerhealthandsafety/equipment-and-innovation/rules-changes/nfl-health-and-safety-related-rules-changes-since-2002>.

In a video game a receiver's block is a formality. In the NFL a crack block is tightly regulated, because a defender hit from the side by a player running at him can't protect himself. The rule (NFL Rule 12, Section 2, Article 6) applies to a player who lines up more than two yards outside the tackle, or comes in motion from the backfield, and then moves back toward the ball: within five yards of the line of scrimmage he may not hit a defender below the waist, nor in the head or neck. It costs 15 yards.¹ A separate rule since 2019 bans the *blindside* block, a forcible hit thrown while the blocker is moving toward or parallel to his own end line,² and from the 2026 season blindside blocks are one of the five fouls in the league's three-strike suspension policy (three in a season means a one-game suspension).³ A legal crack is thrown with the hands and shoulders, from the side, chest-high, and receivers practise it the way linemen practise down blocks.

The jet sweep: a running start at the edge

The **jet sweep** is the purest perimeter play: a receiver in jet motion takes the ball from the quarterback without slowing down and races for the edge on the side he's heading toward. You met the motion and the defense's three answers to it (travel, bump, spin) in 02-06. This section is about the run itself, and about what it does to the linebackers.

Three things make it work:

- **The running start.** The receiver is already at full speed when the ball is snapped; the defenders start from a standstill. The simple model in 02-06 put the gap at about three yards after one second. On a play that is a race to the corner, three yards is the race.
- **The angle.** By the time he has the ball, the runner is already level with the guard on the far side, heading for the edge. The defense's force player has to cover more ground than he does.
- **What it does to everyone else.** Linebackers are taught to read the backfield, and a receiver sprinting across the formation with the ball is the loudest thing in the backfield. Their first steps go with him. Whether or not he gets the ball, the defense's whole second level is now moving sideways, which is exactly what the rest of this chapter takes advantage of.

The mechanics vary. In some offenses the quarterback hands the ball off in front of himself; in others (Kansas City's, for one) he flips it forward a foot or two to a receiver passing in front of him, a "pop" or shovel toss that counts as a forward pass in the record books.⁴ Either way, the runner is gone before the defense has finished its first step.

³The 2026 NFL/NFLPA three-strike policy (hip-drop tackles, horse-collar tackles, launching, hits on a defenseless player and blindside blocks; three in a season means a one-game suspension): <https://www.nbcsports.com/nfl/profootballtalk/rumor-mill/news/nfls-new-three-strike-system-doesnt-apply-until-the-2026-regular-season>.

⁴A pass is forward if the ball is thrown toward the opponent's goal line, however short the distance (NFL Rule 8, Section 1); a flip that travels forward to a receiver crossing in front of the quarterback is therefore a forward pass and is scored as a completion, not a run.

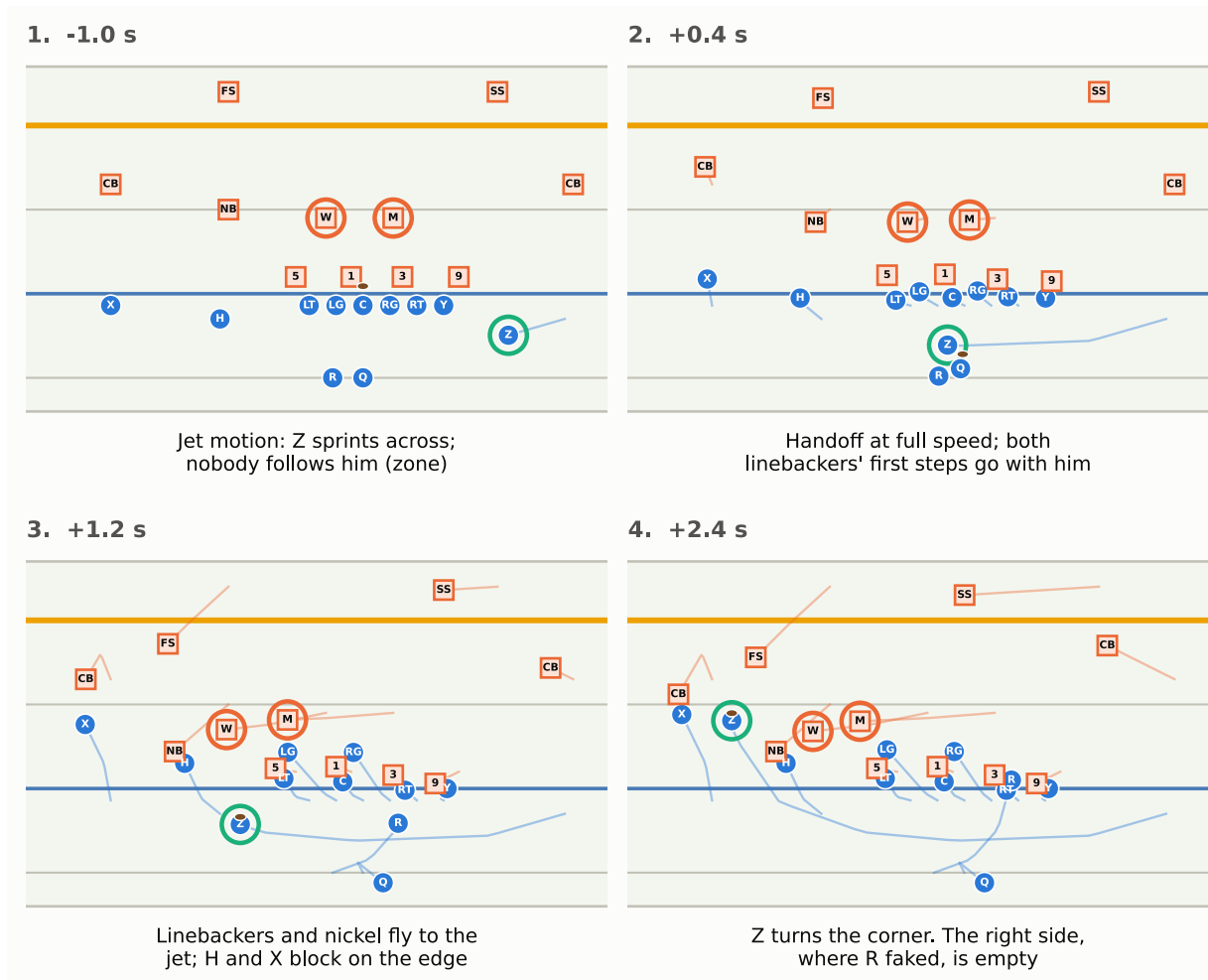


Figure 3: A jet sweep left against a nickel defense playing zone (simulated, BDB-format tracking; faint lines are each player's path so far). 1: Z sprints across in jet motion; nobody follows him, so the defense is in zone. 2: Z takes the handoff at full speed, and both linebackers' first steps go with him. 3: the linebackers (orange rings) and the nickel back run hard to the jet side; H blocks the nickel and X the cornerback, while the line steps left. 4: Z turns the corner with the deep safety coming down to meet him. The right side of the field, where the running back faked a run, is now empty: that is where the plays later in this chapter attack.

Look at the third and fourth panels of Figure 3, not at Z. Five defenders (both linebackers, the nickel back, the cornerback and a safety) are all running toward the left sideline, and the defense's right side has two defensive linemen and a cornerback standing near grass nobody is using. That is the jet sweep's second job. Even when it gains four yards, it teaches the defense that the left sideline is where the danger is, and a defense that has learned that lesson will run there faster next time.

What it gives up. The jet sweep is a high-variance play. When the force player holds the edge, the runner is caught going sideways behind the line, and the loss can be large because he started deep. The exchange happens at full speed between two players moving in different directions, which adds fumble risk. And it uses a receiver as a runner, so every jet sweep is a play on which that receiver isn't running a route. Coaches accept all of that because of what the motion does to the defense when the ball goes somewhere else, which is the subject of the rest of the chapter.

End-arounds and reverses: sending the ball back

If the defense is going to run hard one way, the obvious punishment is to give the ball to someone running the other way. There are two classic ways to do it, and broadcasters mix up their names all the time.

An **end-around** gives the ball to a receiver who comes back behind the line from his spot outside and runs toward the other side of the field. The quarterback hands it to him directly: one exchange. Usually the play starts with a fake the opposite way, so the defense flows toward the fake while the receiver runs away from it. (The name is older than the modern receiver: the "end" in end-around is the end of the line, the position that became the wide receiver and the tight end.)

A **reverse** has two exchanges. The ball first goes to a player running one way, usually a running back on a sweep, and he hands or pitches it to a teammate running the other way. The ball's direction *reverses* behind the line. A double reverse does it again.

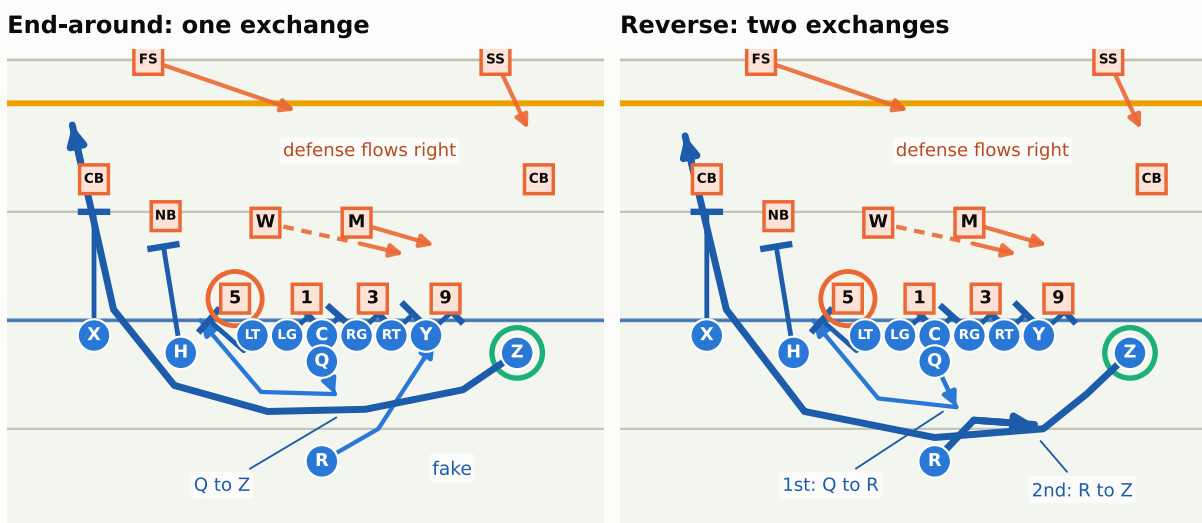


Figure 4: Two ways to send the ball back against a defense flowing right (dashed orange arrows). Left, an end-around: the quarterback fakes a handoff to the running back going right, then hands the ball to Z, who came behind the line from his spot on the right and runs left. One exchange, about four yards deep. Right, a reverse: the quarterback hands off to the back going right, and the back hands the ball to Z coming the other way. Two exchanges, about six yards deep. In both, the play depends on the backside end (ringed): here the left tackle lets him come upfield and then blocks him, and the quarterback leads around the end.

Both plays bet on the same thing: that the defense will chase the first action hard enough to leave the other side empty. Both are beaten by the same defender. On every outside run, the end on the *backside* (the side away from the play) has a simple rule that most defenses teach in some form: don't chase flat down the line; squeeze, stay behind the ball, and watch for anything coming back. Coaches call him the *contain* player on that side, a term 05-01 develops. If he stays home, the reverse runs straight into him six yards behind the line of scrimmage. That is why the drawing gives the left tackle the job of letting that end come upfield and then blocking him, and why the quarterback, who has nothing else to do, leads around the edge.

When you see one, ask two questions. Which defender was responsible for the backside, and did he chase? And how deep was the exchange? A reverse handed off seven yards behind the line has to gain seven yards just to get back to zero. It's a play for a defense that has been caught chasing on film, not a play to run every week.

The Wing-T: an order of football

Most of the ideas in this chapter were organized into a system long before the NFL adopted them. The Wing-T, the formation with a wingback just outside and behind the tight end, was built by Dave Nelson at Maine and then the University of Delaware, where he coached from 1951 to 1965. His assistant Harold "Tubby" Raymond succeeded him and ran it for 36 more seasons, until 2001, winning 300 games, a total few coaches have reached at one school.⁵ Raymond, a Michigan graduate famous for painting a portrait of each of his senior players, and his assistant Ted Kempinski wrote the book on it, and the title says what it is about: *The Delaware Wing-T: An Order of Football*.⁶

The problem Nelson faced was the one every under-dog coach faces: his players were usually not bigger or faster than the other team's. He couldn't win by blocking every defender one-on-one, so he built an offense that made defenders block *themselves*, by running to the wrong place.

The buck sweep

The Wing-T's signature play is the **buck sweep**. Its name tells you its two halves: the fullback fakes the *buck* (an old word for a straight plunge up the middle) and the halfback runs the *sweep*.

⁵University of Delaware, "Blue Hens Rewind: Tubby Raymond wins 300th career game" (Raymond's 300th win came on Nov. 10, 2001, 10-6 over Richmond; he retired at 300-119-3): <https://bluehens.com/news/2020/10/17/football-blue-hens-rewind-tubby-raymond-wins-300th-career-game-in-defensive-battle.aspx>; Nelson's years (1951-1965) and the Wing-T's origins: https://en.wikipedia.org/wiki/Delaware_Fightin%27_Blue_Hens_football; College Football Hall of Fame: <https://cfbhall.com/inductees/tubby-raymond-2003>.

⁶Harold R. "Tubby" Raymond and Ted Kempinski, *The Delaware Wing-T: An Order of Football* (West Nyack, NY: Parker Publishing, 1986).

Buck sweep right

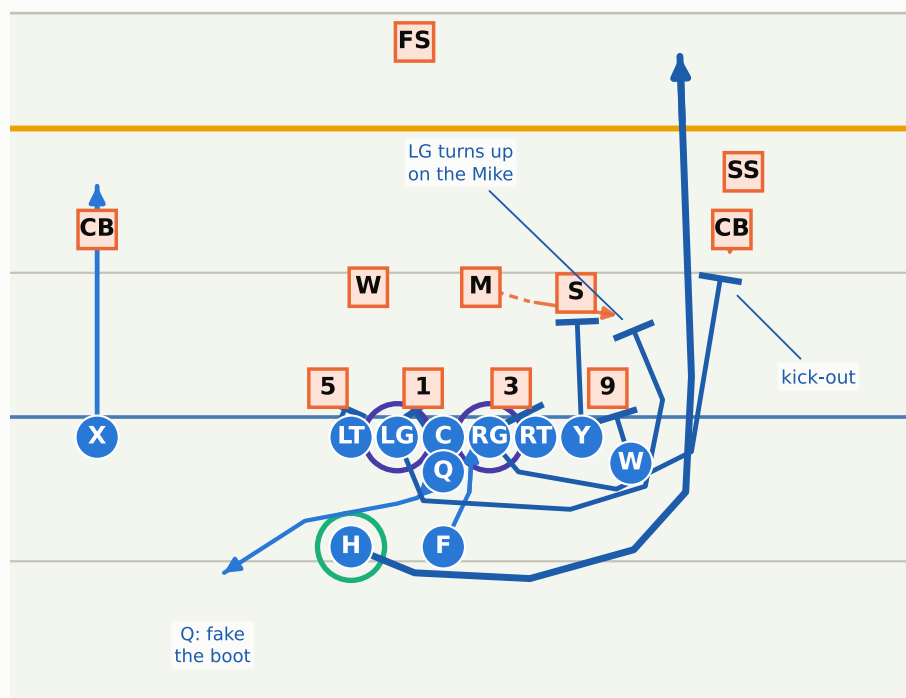


Figure 5: The Wing-T buck sweep, right (wingback W outside the tight end Y; halfback H behind the left tackle; fullback F behind the quarterback; split end X on the left). Everything on the play side blocks down: the wingback on the end, Y on the Sam, the right tackle on the 3-technique. Both guards pull (purple rings): the right guard kicks out the first defender outside the wingback's block (here the cornerback, who comes up to force), and the left guard turns up inside him to block a linebacker. The center blocks back into the hole the left guard left, and the fullback fakes the buck into the gap the right guard left. The halfback takes a reverse-pivot handoff and follows the guards; the quarterback fakes a bootleg the other way.

The buck sweep is power football, the same family as the gap schemes of 03-03: down blocks to wall off the inside, pullers to add blockers at the point of attack. It is also a close relative of the most famous run in NFL history, Vince Lombardi's Packers sweep of the 1960s, which also pulled both guards around a down-blocking line, and an ancestor of today's pin-and-pull.

What makes it a Wing-T play rather than just a sweep is everything that happens in the backfield *before* the ball goes outside. In the first second the defense sees the fullback plunge at the middle, the quarterback spin with his back to the line, the halfback run across, and both guards pull. Then, a beat later, the quarterback rolls the other way without the ball. A linebacker who reads the fullback steps up; one who reads the guards flies outside; one who reads the quarterback drifts backside. Only one of them is right.

Series football: every play is the answer to the last one

That first second is the point. Raymond's Wing-T was built as **series football**: groups of plays that begin with the *same* backfield action and end differently, each one designed to beat the defense's reaction to another. Chris Brown, writing about Raymond's influence, put the idea in one line: "each time a team tries to stop one thing, it opens itself up to something else."⁷ The buck series is the classic example.

⁷Chris B. Brown, "The Making of a Modern-Day Guru," Grantland, January 3, 2014: <https://grantland.com/features/gus-malzahn-offensive-evolution/>.

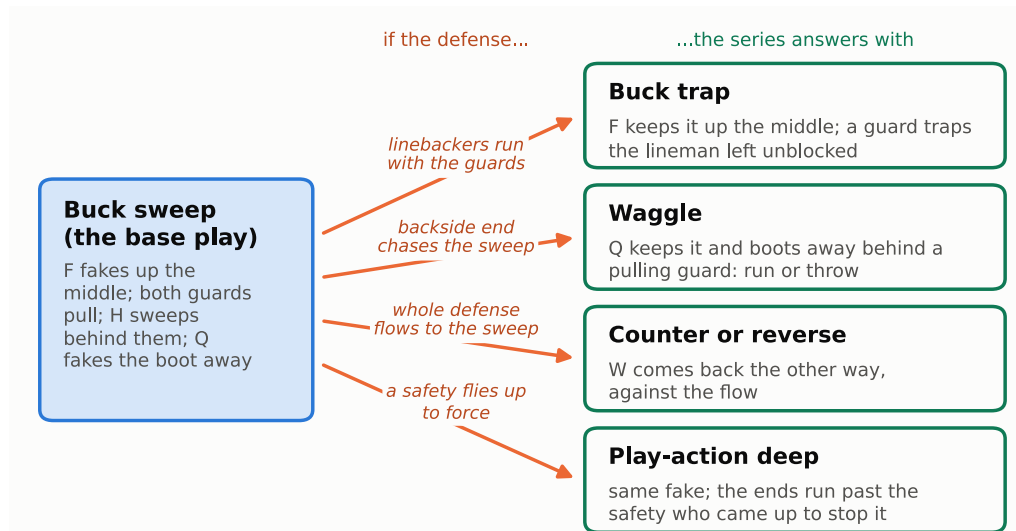


Figure 6: The buck series as a conversation. Every play starts with the same first second (blue box, left). Each orange arrow is something the defense might do to stop the buck sweep, and the box it leads to is the play in the series that punishes that reaction.

The plays are illustrative of the Wing-T's logic; real playbooks have more members and different names.

Read Figure 6 from left to right. If the linebackers learn to fly outside with the pulling guards, the fullback, who faked into the middle on the sweep, keeps the ball this time, and the middle they just vacated is where he runs: the **buck trap**, a trap in which the defensive lineman left unblocked at the snap gets hit from the side by a pulling guard. If the backside end chases the sweep, the quarterback keeps the ball and rolls out behind him. If the whole defense flows, the wingback comes back the other way on a counter or reverse. If a safety races up to make the tackle on the sweep, the receivers run past him on play-action (a fake handoff followed by a pass).

None of those plays is clever on its own. A trap is a trap. What makes the series work is that each play is *indistinguishable from the others for the first second*, and that the defender's correct reaction to one is his wrong reaction to another. The defense can't stop all of them by being fast or by being patient; it has to be right, play after play, about a fake it can't see through. That is what "an order of football" means: plays called in a deliberate order, each one asking the defense a question the previous one made it answer.

The waggle

One member of the series reaches straight into the modern NFL. The **waggle** is the Wing-T's play-action bootleg: the quarterback fakes the buck sweep (or another run) one way and rolls out the other, behind a guard who pulls to that side to protect him, with receivers spread at three depths on that side.

If that picture looks familiar, it should. Replace the Wing-T backfield with a single back running outside zone and you have the bootleg that Mike Shanahan's and Kyle Shanahan's offenses, and Sean McVay's, built on (taught in full in Play-Action, Bootlegs, and Screens, where the version with no pulling guard at all, the *naked* bootleg, gets its own treatment). The waggle's idea, fake hard one way and put the quarterback on the edge the other way with three receivers at three depths, survived when the formation that invented it disappeared.

You will almost never see a true Wing-T in the NFL. You will see its children every Sunday.

⁸Daniel Rymer, “Examining the jet sweep revolution of the 2018 NFL season,” Pro Football Focus, May 13, 2019 (174 jet sweep runs for 5.1 yards a carry in 2017; 309 for 6.1 in 2018; the Rams 48, including playoffs): <https://www.pff.com/news/pro-examining-the-jet-sweep-revolution-of-the-2018-nfl-season>.

Gus Malzahn's Auburn. The clearest line from Delaware to modern football runs through Gus Malzahn. As a young high school coach in Arkansas in the early 1990s, he taught himself offense from Raymond and Kemp'ski's book; he soon moved to an up-tempo shotgun spread, but kept the Wing-T's plays and its series logic, and at Auburn he ran the buck sweep from the shotgun with pulling linemen and a receiver faking an end-around the other way.⁹ His 2013 team averaged more than 300 rushing yards a game, the first SEC team to do it in almost thirty years.

The NFL's misdirection revival. When the McVay Rams and Kyle Shanahan's 49ers made jet motion part of almost every play in the late 2010s, they were doing what Raymond did with his backfield: one picture, several plays. Shanahan went a step further with Deebo Samuel, a receiver he lined up in the backfield, put in motion and handed the ball on sweeps, reverses and runs between the tackles, so that "which player is the runner" became one more question the defense had to get right. Andy Reid's Chiefs used jet and orbit motion to set up flips, shovels and screens back the other way. The formations are modern; the logic is the Wing-T's.

Eye candy: half a second is a gap

Here is the payoff from Figure 3. Once the defense has learned to fly at jet motion, the motion itself becomes a weapon even when the ball goes somewhere else. Coaches call this **eye candy**: motion and fakes by players who won't get the ball, designed to pull defenders' eyes and first steps the wrong way.

How much can a fake buy? Not much, and that's enough. A linebacker who takes one hard step toward a jet fake and then has to come back has lost about half a second. Linebackers move at something like 5 to 7 yards a second when they're trying to fill a hole; half a second is three yards. A hole on an inside run is a yard or two wide and stays open for less than a second. A linebacker three yards out of place doesn't fill it.

⁹Brown, "The Making of a Modern-Day Guru" (Malzahn learning the offense from Raymond and Kemp'ski's book as a high school coach; adapting the buck sweep to the shotgun; Auburn 2013 as the first SEC team in almost 30 years to average more than 300 rushing yards a game). The Grantland article is the source for this paragraph; see also Malzahn's own *The Hurry-Up, No-Huddle: An Offensive Philosophy* (2003).

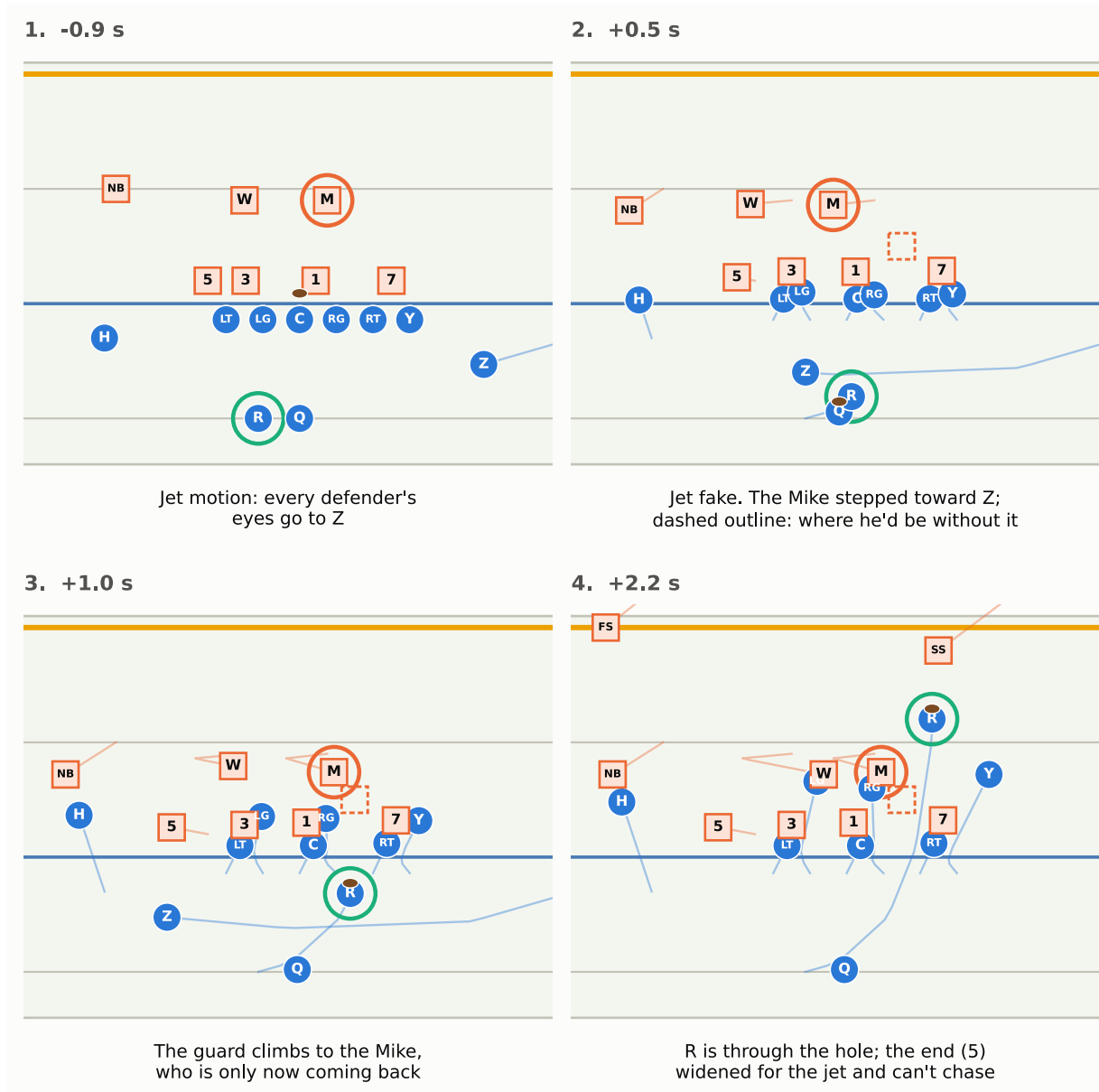


Figure 8: Eye candy: a jet fake left, an inside zone run right (simulated, BDB-format tracking). The dashed outline is where the Mike linebacker would be at the same moment if there had been no jet motion and he had simply read the run (a second simulation of the same play without the motion). 1: Z in jet motion; every defender's eyes go to him. 2: at half a second the quarterback has faked to Z and is handing to R. The Mike (ringed) took his first step toward the jet; without the fake he would already be filling the hole. 3: the right guard climbs to the Mike, who is only now coming back. 4: the back is through the hole. The backside end (5) also widened for the jet, so he is too far away to chase the back from behind.

Two defenders were taken out of Figure 8 without being blocked. The Mike linebacker's first step went the wrong way, so the guard who climbed to him got there in time. And the backside end, who on a normal inside zone run is the unblocked defender who can chase the back down from behind (03-02 spent a whole section on him), widened to protect against the jet sweep, so he was out of the play. One receiver running across the formation did the work of two blockers.

That is the trade eye candy makes. The receiver in motion isn't blocking anybody, so the offense starts a man short at the point of attack, and the motion has to hold *more* than one defender for the play to come out ahead. It usually does only when the defense believes the motion. **Eye candy is only as good as the last time the ball actually went there.** Robert Woods, one of the 2018 Rams' two main jet-sweep runners, carried the ball only 19 times in the regular season,¹⁰ but the Rams faked the jet constantly; the defense had to treat every fake as real because some of them weren't.

¹⁰nflverse play-by-play, 2018 regular season: Robert Woods 19 carries, 157 yards, 1 touchdown (course calculation; matches Pro Football Reference's season line). PFF's charting, which also counts jet plays scored as passes, credits Woods with 16 jet sweeps including the playoffs (Rymer, above).

Jet motion is the most common kind of eye candy, not the only one. An orbit motion behind the quarterback, a running back faking a sweep one way while the ball goes the other, a tight end running across the formation as though he's going to block, the quarterback carrying out a bootleg fake after he hands the ball off: all of them are designed to make a linebacker's eyes and feet do something for a fraction of a second that they shouldn't.

Common misconception

"The jet motion was wasted; the guy didn't even get the ball." On most plays with jet motion the ball goes to someone else, by design. Judge the motion by what the defenders near it did. If two linebackers took a step toward the motion man and the run went through the gap one of them left, the motion was the most important "block" on the play.

How defenses fight back

None of this is free for the offense, because good defenses train against every trick in this chapter.

- **Keep the backside honest.** The backside end and the backside cornerback have rules on outside runs, and the first rule is "don't chase until the ball has crossed the formation." That kills reverses and end-arounds and makes bootlegs and waggles harder, at the price of one fewer defender pursuing the play that really does go outside.
- **Answer the crack.** When a receiver cracks inside, the cornerback over him yells "crack!" and steps up to replace the man who was cracked as the force player. The puller then has nobody to block, and the back is running at a cornerback who is already outside him.
- **Spin to the motion.** Rotating a safety down toward the jet side at the snap (the *spin* of 02-06) puts an extra defender at the edge, so the linebackers don't have to chase. The offense's counter is to run the other way, into the side the safety left.
- **Read your keys, not the motion.** Linebackers are taught to read the linemen in front of them (their *keys*, which 05-04 covers), not the receiver in motion. A guard stepping right means the run is going right, whatever the jet man is doing. The problem is that the jet man is going *very fast*, and a linebacker who ignores him entirely gets beaten to the edge.

That last point is the defense's version of the dilemma in Figure 1. Every answer that stops misdirection makes the defense slower to the perimeter, and every answer that stops the perimeter run makes it more vulnerable to misdirection. The full catalog of run fits, including who forces and who spills, is in Run Fits.

Constraint plays: why you can't just run it more

Here is a puzzle in the numbers. When an NFL wide receiver carries the ball on a run, almost always on a jet sweep, an end-around or a reverse, the play works much better on average than a running back's carry. So why do teams hand it to receivers less than once a game?

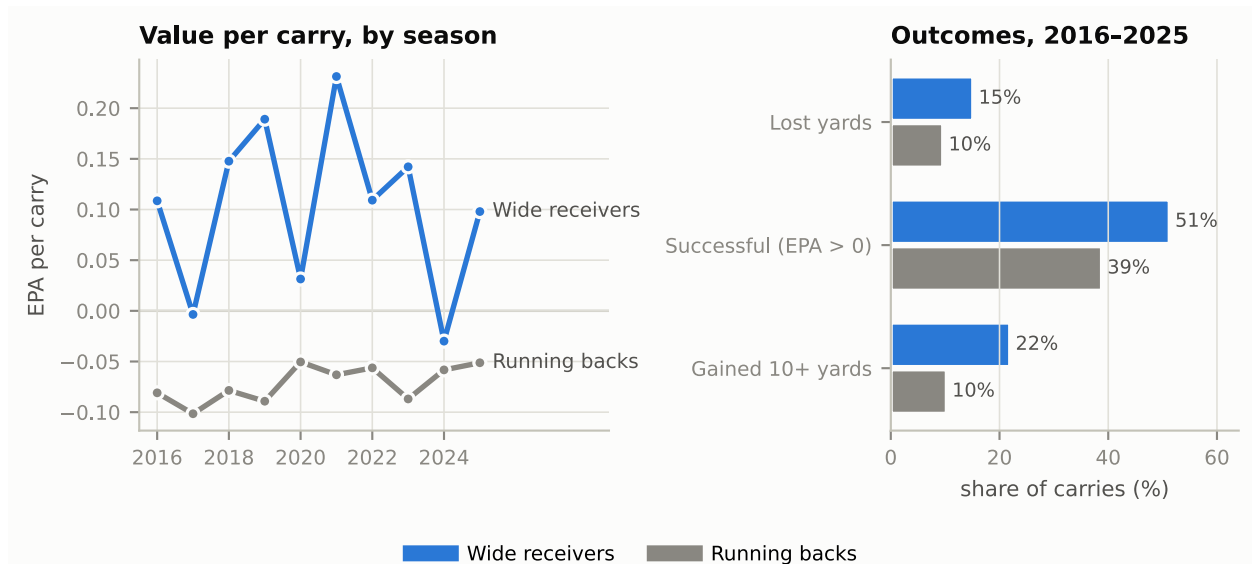


Figure 9: Wide receivers as runners, NFL regular seasons 2016–2025. Left: expected points added (EPA) per carry by the ball carrier’s position. A running back’s average carry loses a little expected value every season; a wide receiver’s is worth more than a running back’s in every season, and is positive in eight of the ten. Right: the shape of the outcomes, all ten seasons pooled. Receiver runs are more often successful and twice as often go for 10 or more yards, but they also lose yards more often. League-wide there were only 0.6 to 0.9 receiver carries per team per game. Runs only (jet flips scored as forward passes are not included); kneel-downs removed; position from nflverse season rosters. Data: nflverse play-by-play via nflreadpy; course calculation.

The numbers in Figure 9 are striking. Across ten seasons, a running back’s carry was worth about -0.07 expected points on average and succeeded (gained expected points) 39% of the time. A receiver’s carry was worth about $+0.10$ and succeeded 51% of the time, and it gained ten or more yards on 22% of carries, more than twice the running backs’ rate. If you have read 13-02, you’ll know that a gap of 0.17 expected points a play is enormous. It’s roughly the gap between a good passing play and a bad run.

But notice the other bar: receivers *lose* yards on 15% of their carries, against 10% for backs. That is the jet sweep and reverse caught behind the line. And notice the volume: well under one receiver carry per team per game, every season. Teams aren’t ignoring free yards. The yards come *from* the rarity.

Coaches call plays like these **constraint plays**, plays whose main job is to punish the defense for overplaying the offense’s *base* plays, the bread-and-butter runs and passes it calls most. The idea is developed fully in Constraint Theory, but the shape is already visible here. A jet sweep gains a lot when the defense isn’t ready for it, and the defense isn’t ready for it because the offense mostly runs something else. Run it twenty times a game and the defense would stop playing the inside run and start sitting on the edge, the jet sweep would turn into a two-yard loss, and the inside run would get worse too, because the linebackers would no longer be taking those hard first steps toward the motion. The value of a misdirection play lives partly in the plays it *isn’t*.

The same caution applies when you read the chart. Some of the receivers’ edge is selection: coaches call these plays when the film says the defense is vulnerable to them, and the defense often isn’t expecting a receiver run at all. That is not a flaw in the numbers; it’s the point. A constraint play’s average looks great *because* it’s called at the right moments.

! Why it exists

Misdirection exists because pursuit works. A defense that runs to the ball fast and together is the best answer to almost every run in football, so the offense needs plays that make running fast to the ball a mistake: plays that go back the other way (reverses, counters, waggles) and motion that makes the defense run to the wrong place (eye candy). The price is risk and blockers: misdirection plays lose yards more often, take longer to develop, and use players as decoys who could have been blocking. They work only in the right dose, which is why the best offenses build them into series, so the base play and its answers look the same until it’s too late.

Film room

i Film room: Delaware, 1966-2001 (Tubby Raymond's Wing-T)

There is no tracking data for Raymond's Delaware teams, and you don't need it. If you can find old Blue Hens film, watch the backfield and not the ball for the first second of any play: the fullback plunging, the quarterback spinning with his back to the line, the halfback crossing, both guards pulling. Then watch the linebackers. On the buck sweep they are a step slow to the outside because the fullback's fake made them look inside first; on the trap and the waggle they are a step out of the middle because the sweep has been killing them. The plays changed little for decades. Delaware kept winning with them because the defense's problem never went away: a defense has to react to *something*, and everything it can react to is a fake on most plays.

i Film room: Auburn 2013 (Gus Malzahn)

Malzahn's 2013 Auburn team, led by quarterback Nick Marshall and running back Tre Mason, ran for more than 300 yards a game and reached the national championship game. What to notice: the buck sweep from the shotgun. The guards pull exactly as in Figure 5, but the quarterback is five yards deep and the "halfback" is lined up beside him. A receiver often runs across the backfield the other way, faking the end-around, the Wing-T's buck fake moved to a different player. And the series is there too: the same look produces the sweep, an inside run and a quarterback keep.

i Film room: Los Angeles Rams 2018 (Sean McVay)

The 2018 Rams made the jet sweep an NFL staple: PFF counted 48 of them, playoffs included, twice as many as any other team.¹¹ In the nflverse play-by-play, Rams wide receivers carried the ball 35 times in the regular season, and those carries averaged about +0.44 expected points, among the best in the league that year (course calculation). The two counts differ because a charting service tags the *play*, while play-by-play records who carried the ball, and some jet plays are scored as passes. What to notice: the jet sweep is blocked like outside zone, the whole line stepping toward the sweep, so for the first second it looks exactly like the Rams' base run going the same direction. Then watch the next play with the same motion: half the time the ball goes the other way, or the quarterback fakes it and throws.

i Film room: San Francisco 49ers, Super Bowl LIV (2019 season)

Deebo Samuel was a rookie receiver in 2019 who ran the ball 14 times for 159 yards and three touchdowns in the regular season.¹² In Super Bowl LIV against Kansas City he carried three times for 53 yards, two of them gains of 32 and 14 yards around the ends.¹³ What to notice: the 49ers didn't hide that Samuel might get the ball. They moved him around so the defense had to decide, on every snap, whether the receiver crossing the formation was a decoy. When linebackers stayed home to stop the inside run, he got the ball at the edge; when they chased him, the 49ers ran Raheem Mostert inside. By 2021, as 02-06 showed, Samuel was effectively a second running back.

¹¹Rymer, PFF (as above): the Rams' 48 jet sweeps (2.3 a game, 6.8 yards a carry, three touchdowns), with the Patriots and Chargers next at 24. The nflverse figures are a course calculation: regular-season runs by players listed as wide receivers on the Rams' 2018 roster.

¹²nflverse play-by-play, 2019 regular season: Deebo Samuel 14 carries, 159 yards, 3 touchdowns (course calculation).

¹³nflverse play-by-play, Super Bowl LIV (Feb. 2, 2020): Samuel carries of 32, 7 and 14 yards (course calculation).

Watch for it on Sunday

Watch for it

Watch the second level. On every play with jet or orbit motion:

Before the snap

- Find the motion man and the side he's heading to. Does anyone travel with him (man coverage), does everyone slide a spot (zone), or does a safety spin down toward his side?
- Find the two linebackers and the backside end. These are the players whose reaction you'll grade.

At the snap: forget the ball for one second

- Watch the linebackers' eyes and feet. Did they take a step toward the motion? Did they keep going?
- Did the backside end squeeze down the line, or stay home?

After the play

- Where did the ball go? If it went *with* the motion: did the force player hold the edge, or did the runner beat him? Was there a crack block on the edge?
- If it went *away* from the motion: were the linebackers late because they chased? That's eye candy working.
- If the defense has been chasing all game, start expecting the play that goes back the other way: a counter, a reverse, a bootleg.

For your prediction log (01-06): on every jet motion, write **G** (the motion man gets it) or **F** (fake) before the snap, then **C** if the linebackers chased, **H** if they held. After a game you'll have a small table of what this defense does with motion, which is exactly what offensive coordinators study during the week.

Predict the play

💡 Predict the play

Drill 1. First-and-10, ball on the offense's own 35. The offense has run three jet sweeps to the left today, for 9, 12 and 7 yards. Here Z goes in jet motion again, right to left. This time, as he crosses, the strong safety (SS, ringed) comes down hard toward the left edge and the free safety slides to the middle of the field. The two linebackers (W and M) don't move. Should the quarterback give Z the jet sweep? If not, where would you expect the ball to go?

Drill 1: jet motion, safety spins down

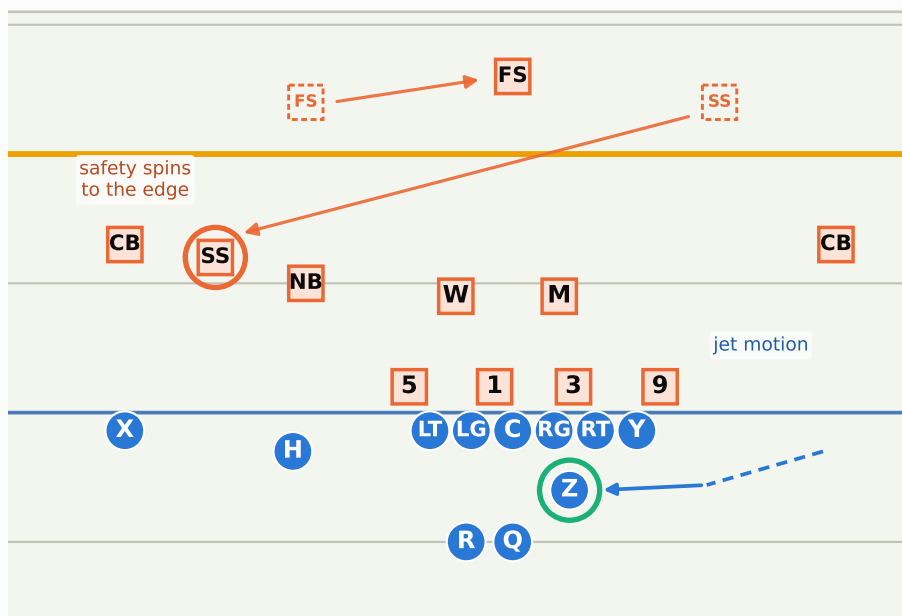


Figure 10: Drill 1, at the snap. Z (green ring) is in jet motion from the right (dashed blue path). The strong safety (SS, ringed) has rotated down toward the left edge and the free safety has moved to the deep middle; hollow outlines show where they started. The linebackers (W, M) haven't moved.

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play

Drill 2. Second-and-4. The offense is in the Wing-T and has run the buck sweep right twice in a row from this look. Both times, at the snap, the Mike and Will linebackers (ringed) ran hard to the right with the pulling guards (their paths from those plays are shown in orange). The offense lines up in the same formation again. What does series football call now, and why?

Drill 2: same look, third time

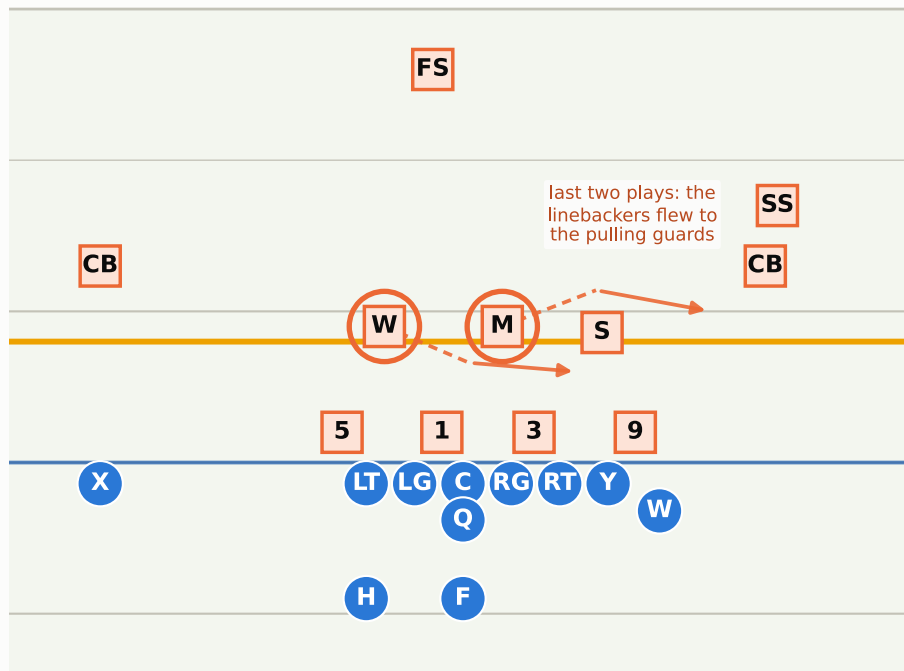


Figure 11: Drill 2: the Wing-T look again (wingback W outside the tight end on the right, halfback H behind the left tackle, fullback F behind the quarterback). Dashed orange arrows show where the two linebackers (ringed) ran on the last two buck sweeps, as soon as they saw the guards pull.

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play

Drill 3. First-and-10. The offense is under center with two tight ends, and Z is lined up unusually tight on the right, about eight yards from the ball and a yard off the line, *inside* the cornerback. The running back stands seven yards deep, slightly offset away from Z. The Sam linebacker is lined up outside the tight end, between the tight end and Z. If this is the run it looks like, what is it, which defender will Z block, who blocks the cornerback, and what should the cornerback do about it?

Drill 3: condensed split, back deep

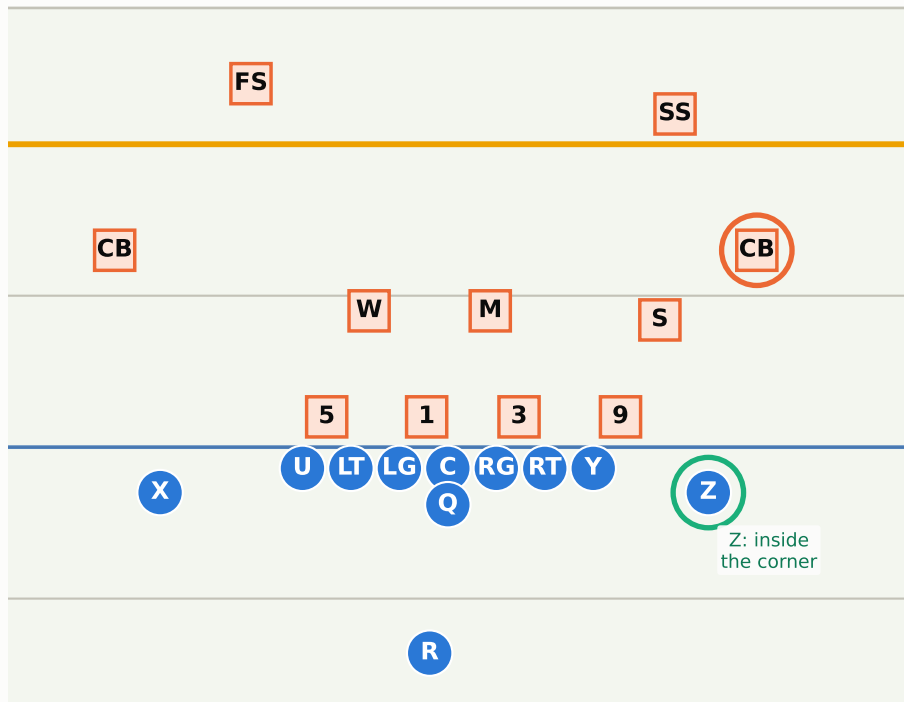


Figure 12: Drill 3: a condensed two-tight-end set. Z (green ring) is about $8\frac{1}{2}$ yards from the ball, a yard off the line and inside the cornerback (CB, ringed), who is lined up a yard and a half outside him. The Sam linebacker (S) is between the tight end and Z. The running back (R) is about seven yards deep.

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- Outside runs are a **race** between the ball and the defense's pursuit. The defense wins it with a **force** player who turns the ball back inside and everyone else pursuing fast. Perimeter plays punish a slow defense; misdirection punishes a fast one, and each family needs the other.
- Two blocks make outside runs work: the **crack block**, a receiver or wing hitting an inside defender from the side (above the waist, by rule), and the **seal block**, walling a defender off rather than moving him. Together with a puller replacing the cracking receiver, they make the **crack-toss**.
- The **jet sweep** turns jet motion's running start into a race the defense starts a step behind, and pulls the linebackers sideways even when the ball goes elsewhere. **End-arounds** (one exchange) and **reverses** (two) send the ball back against the flow and live or die on the backside end.
- The **Wing-T** organized all of this as **series football**: the **buck sweep** and the plays that look identical for one second and punish each reaction to it (the trap, the **waggle**, the counter). Its children are the NFL's jet sweeps, buck sweeps from the shotgun and bootlegs.
- **Eye candy** is motion or a fake that holds defenders without blocking them. Half a second is about three yards, which is a gap. It works only as long as the motion is sometimes real.

- Receiver runs are rare (under one per team-game) and, on average, much more valuable than running back carries, with more lost-yardage plays. They are **constraint plays**: their value comes from the base plays they punish, which is why you can't simply run them more.

Connections

Builds on: Gap Schemes (pullers, kick-outs, logs, trap, toss and pin-and-pull), Motion and Shifts (jet, fly and orbit motion, and the defense's travel, bump and spin), Zone Running (outside zone, the backside end, the bootleg), Classic Formations (the Wing-T formation) and Spread and Modern Formations (condensed splits and the first look at the crack).

Sets up:

- Play-Action, Bootlegs, and Screens: the waggle's descendants, the bootleg and the naked bootleg, and the play-action passes that live off jet motion.
- Run-Pass Options: jet motion and quick screens attached to inside runs, the modern way to put a defender in a bind on every snap.
- Linebackers and the Second Level and Run Fits: keys, flow, force and spill, the defense's side of everything in this chapter.
- Constraint Theory and Sequencing: base plays, constraint plays and why the best offenses call them in a deliberate order.
- The College Laboratory: the Wing-T, the wishbone and the spread option, and how college football keeps feeding the NFL ideas.

See also (case studies): The Shanahan Tree (wide zone, boots and jet motion as one system) and Andy Reid's Chiefs (motion, shovels and the creative run game).

If you want to go deeper: Raymond and Kempinski's *The Delaware Wing-T: An Order of Football* is still the clearest statement of series football. Chris B. Brown's *The Art of Smart Football* and his Grantland piece on Gus Malzahn trace the Wing-T into the modern spread. For the NFL version, watch a 2018 Rams or 2019–2021 49ers game on the All-22 and count the linebackers' first steps on every jet motion.

Go deeper

From diagrams to real tracking data. The two animations in this chapter are gridiron simulations in Big Data Bowl (BDB) coordinates, so real tracking data drops into the same drawing code. The 2025 Big Data Bowl release (2022 season, weeks 1–9) flags, for every player on every play, whether he was in motion at the snap, and records who carried the ball, so you can pull every run by a wide receiver who was moving at the snap (most of them jet sweeps) and watch the linebackers' first steps for yourself. The data is hosted on Kaggle and needs a free login, so this code is not run when the book is built.

Answers to Predict the play

Answer 1.

No jet sweep, probably a run the other way (or play-action). The safety's spin has done the job the linebackers usually do: the defense now has an extra player at the left edge before the play starts, so the jet sweep would run into the cornerback, the nickel *and* a safety coming downhill. Meanwhile the linebackers stayed home, so a fake to the jet won't move them either. The space is on the **right**, the side the strong safety just left: one deep safety in the middle, and nobody near the right edge except a cornerback. Expect the running back on an outside run to the right, a run with the tight end leading on that side, or a play-action pass behind where the safety used to be.

Compare this with the jet drill in 02-06, where the linebackers didn't move and *nobody* replaced them at the edge, so the jet was the right answer. Same linebackers, different safety, opposite call. Many offenses let the quarterback choose between the jet and the inside run at the snap from exactly this read.

Answer 2.

The buck trap: the fullback keeps the ball up the middle. The first second will look the same as the sweep: quarterback reverse-pivots, halfback runs across, the fullback heads for the middle. But this time the fullback has the ball, and the linemen block for an inside run. A defensive lineman is left unblocked at the snap and a guard pulls a short distance to trap him (hit him from the side), so the fullback runs through the gap where that lineman *was*, straight into the space the two linebackers are about to vacate by chasing the guards. The more the linebackers trust what they saw on the last two plays, the bigger the hole.

Why not something else? The waggle punishes the *backside end* for chasing, and the drill told you about the linebackers. A reverse or counter punishes the *whole defense* flowing, and here only the second level is flowing hard. Series football means picking the

member of the series that beats the specific reaction you saw. On second-and-4 an inside run that only needs four yards is also exactly the right risk.

Answer 3.

A crack-toss to the right, as in Figure 2. The tells: a receiver lined up *inside* the cornerback (a wide receiver has no reason to start there on a run unless he is going to block back inside), a tight end on the same side to seal the end, and a back deep enough to take a pitch and get outside. **Z cracks the Sam**, the force player he is already lined up beside. **A pulling lineman, often the right tackle, replaces him on the cornerback.**

The cornerback's answer: watch Z, not the backfield. The moment Z turns inside, he yells "crack!" and comes up hard to become the force player himself, *outside* the ball, ideally before the pulling tackle reaches him. That turns the back inside, toward the pursuit. If the corner gets caught watching the backfield, he gets blocked by a 310-pound tackle in space. The offense's counter to a corner who crack-replaces aggressively is to fake the crack and throw over him: Z steps inside, the corner jumps up, and Z turns upfield on a route. One alignment, two plays, and the corner has to be right about which.